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UNSTEADY AIRFOILS[1]

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INTRODUCTION

Flow over streamlined lifting surfaces has been a major topic of fluid mechanics during most of the twentieth century. For simplicity, unsteady effects have often been ignored. However, most modern fluid-dynamic devices encounter or generate unsteady flow, whether intentional or not, for at least part of their operating conditions. In this review, we focus on the role of unsteady effects in an important class of flow problems, namely, two-dimensional oscillating airfoils, and on the advances that have been made within the past decade toward understanding these special and challenging flows.

Studies of unsteady-airfoil flows have been motivated mostly by efforts to avoid or reduce such undesirable effects as flutter, vibrations, buffeting, gust response, and dynamic stall. This requires predicting the magnitude and phase (or time lag) of the unsteady fluid-dynamic loads on thin lifting surfaces. Some attention has also been given to potentially beneficial effects of unsteadiness, such as the propulsive efficiency of flapping motion, controlled periodic vortex generation, and stall delay, and to improving the performance of turbomachinery, helicopter rotors, and wind turbines by controlling the unsteady forces in some optimum way. Most of these studies and applications concern either periodic motion of an airfoil in a uniform stream or periodic fluctuations in the approaching flow. We follow this pattern in this review, to the exclusion of impulsive or other nonperiodic motions. The scope is further limited to two-dimensional flows past streamlined shapes, notwithstanding the practical importance of three-dimensional effects and bluff-body configurations.

Our plan is first to explain and illustrate some basic concepts and unsteady phenomena by means of thin-airfoil theory for an inviscid incom-

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pressible fluid. This background is a valuable framework against which we may then view recent and current research on the various complicating effects of compressibility and viscosity, with particular emphasis on unsteady transonic flow and dynamic stall. We conclude with a discussion of probable developments in the future.

A MODEL PROBLEM—THE OSCILLATING THIN AIRFOIL

Many important features of unsteady-airfoil behavior can be described by linearized thin-airfoil theory. The fluid-dynamic pressure forces acting on a thin lifting surface inclined at a small angle relative to the approaching flow are proportional to the effective angle of attack and to the square of the speed of the flow. If either the body or the flow fluctuates, so do the circulation and the pressure distribution; and each change in circulation around spanwise sections of the body is accompanied by the shedding of free vorticity from the trailing-edge region into the wake. This time-dependent vortical wake is an important distinguishing feature of unsteady airfoils.

Among the best known and most enlightening analyses of this class of problems are those by Theodorsen (1935) and von Kármán & Sears (1938), who considered a thin flat plate and a trailing flat wake of vorticity in an incompressible fluid. For periodic oscillations of the airfoil or the flow, the fluid-dynamic phenomena are characterized by a nondimensional frequency parameter $k = \omega c / 2U_\infty$, and the solution can be expressed in terms of combinations of standard Bessel functions whose argument is k. Here ω is the circular frequency of the oscillation, c is the chord of the airfoil, and U_∞ is the mean free-stream velocity. We examine several illustrative results for inviscid flow in the following paragraphs.

Sinusoidal Oscillations in Pitch

Our reference problem for this review is a flat-plate airfoil oscillating sinusoidally in pitch about an axis located at $X = A$, where the airfoil is defined by $0 \leq X \leq 1$ and the angle of attack is $\alpha = \alpha_1 \mathcal{R}e^{i\omega t}$. Theodorsen (1935) divided the flow into two components. The first is a "noncirculatory" component of sources and sinks, which satisfies the boundary conditions on the oscillating plate and which therefore includes the apparent-mass effects. The second is a "circulatory" component, which includes bound vortices and wake vortices and which is matched to the noncirculatory component at the trailing edge in such a manner as to enforce the Kutta condition of nonsingular flow there. We can distill his results for

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the nondimensional pressure distribution, lift, and pitching moment about the axis of rotation into the following form:

$$C_P = -2\alpha \sqrt{\frac{1-X}{X}} [f_1 + ig_1], \quad (1)$$

$$C_L = 2\pi\alpha [f_2 + ig_2], \quad (2)$$

$$C_M = 2\pi\alpha \left(A - \frac{1}{4}\right) [f_3 + ig_3]. \quad (3)$$

The terms outside the square brackets are immediately recognized as the steady solution for a flat plate at instantaneous incidence α. All of the unsteady effects are included in the functions $f_n + ig_n$, which contain the Bessel functions and which depend only on k, A, and in the case of the pressure, X. Details of the unsteady solution can be found in Theodorsen (1935), Postel & Leppert (1948), Fung (1969), and McCroskey (1973). The real parts of Equations (1)–(3), represented by f_n, designate the "in-phase" components of C_P, C_L, and C_M that follow directly the motion of the airfoil; and the imaginary parts, g_n, are the "quadrature" components that lag or lead the motion by $\omega t = \pi/2$. Alternatively, the airloads are sometimes expressed in terms of the modulus $(f_n^2 + g_n^2)^{1/2}$ and phase angle $\phi = \tan^{-1}(g_n/f_n)$, as shown in Figure 1. In the quasi-steady limit of $k \to 0$, the solutions for f_n approach unity as $(1 - \pi k/2)$, and g_n vanishes as $k \ln k$. This logarithmic behavior of g_n is unique to the two-dimensional case, however.

The important point here is that relatively simple, known functions modify the classical steady-state solution for a thin airfoil when the incidence changes at a finite rate. Equations (1)–(3) give explicitly the relative amplitude and phase of the airloads with respect to the airfoil motion. For example, the solid line in Figure 1 shows the locus of the solution for C_L/α at various values of reduced frequency for the important case of rotation about the static aerodynamic center, $A = 1/4$. For this case, the real part of the lift function f_2 decreases monotonically with increasing k, whereas the imaginary part g_2 changes sign at $k \simeq 0.15$ and increases thereafter. The behavior of both f_2 and g_2 is dominated by noncirculatory terms at high frequency (see Fung 1969).

The pitching moment for $A = 1/4$ is given by $C_M/2\pi\alpha = (3/32)k^2 - i(k/4)$, and the fact that the imaginary part is always negative is significant. This means that C_M always lags the airfoil motion and that the aerodynamic damping in pitch, $-\oint C_M d\alpha$, tends to suppress torsional oscillations at all frequencies. In subsequent sections, we examine cases in which $\mathcal{I}_m(C_M)$ can become positive, leading to the possibility of flutter in pure pitching motion. Normally such behavior is most likely to occur when

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shock waves or separation introduce strong nonlinearities in the flow that violate the assumptions of classical thin-airfoil theory.

Other Types of Motion

Analogous to the preceding example, linear theory has been applied to other types of motion that occur in practical applications. For incompressible fluids, well-known solutions exist for translational oscillations normal to the flight path (plunging motion), longitudinal oscillations, vertical and horizontal gusts, step changes in incidence, oscillating control surfaces, and multiple airfoils (cascades). Lists of references on these important topics can be found in Carta (1978), Ashley (1977), and McCroskey

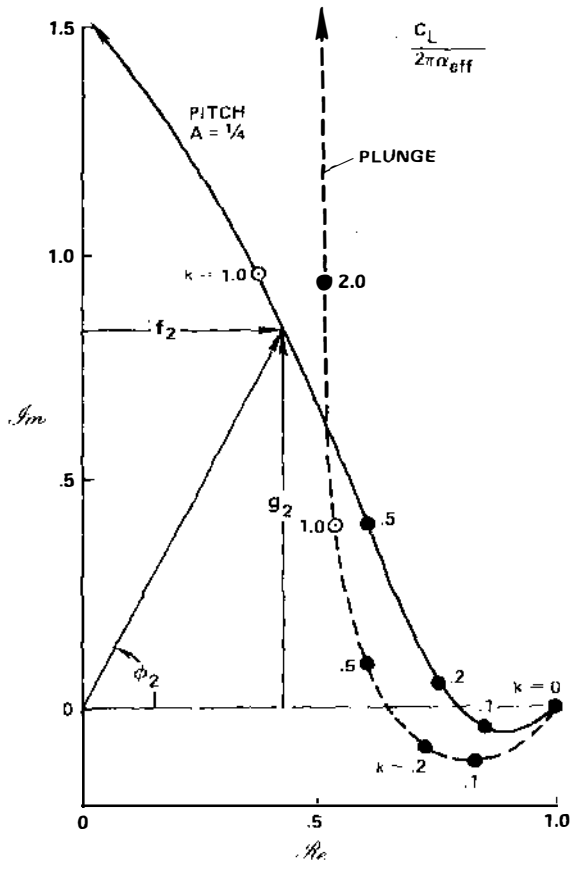


Figure 1 Loci of the real and imaginary components of unsteady lift on an oscillating flat plate as a function of reduced frequency.

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(1977). Excluding the case of oscillating control surfaces for the moment, we can express the pressure distribution, lift, and moment for any of these motions in the general form of Equations (1)–(3). That is, the unsteady effects can always be lumped into unsteady factors $f_n + ig_n$ that multiply the classical steady-state solutions.

VERTICAL AIRFOIL OSCILLATIONS In this case the periodic airfoil displacement $h = h_1 \mathcal{R}e^{i\omega t}$ creates an effective incidence $\alpha_{\text{eff}} = -h/U_\infty$, and this quantity is used to normalize C_L in Figure 1. The unsteady lift behavior with respect to α_{eff} generally resembles the pitching case, but quantitative differences can be seen as k increases.

The lift always lags the plunging *displacement*; therefore, as in the previous case of pure pitching motion, single-degree-of-freedom flutter does not occur. However, *combined plunging and pitching* can produce interactions and phasings between the two degrees of freedom that extract energy from the air stream. This occurs in practice as classical bending-torsion flutter of flexible aircraft wings, which is one of the principal topics of aeroelasticity (see Garrick & Reed 1981).

OSCILLATING CONTROL SURFACES The sinusoidal deflection $\delta = \delta_1 \mathcal{R}e^{i\omega t}$ of a plain flap on a flat plate produces a pressure distribution that consists of the sum of three parts. The first has the same basic form as Equation (1), containing unsteady functions $f_4 + ig_4$ and an inverse square-root singularity with respect to X at the leading edge. The second is algebraically rather complicated (see Postel & Leppert 1948), and contains a logarithmic singularity at the hinge point X_c, where the dominant behavior is given by

$$C_p(X_c) \to \frac{2\delta}{\pi} \ln(X - X_c)^2[f_5 + ig_5] \,. \tag{4}$$

The quasi-steady behavior is again given by $f_4 = f_5 \to 1$ and $g_4 = g_5 \to 0$. The third part, which vanishes as $k \to 0$, is nonsingular and it generally tends to reduce the amplitude of the oscillating pressure caused by the flap motion as k increases. This is illustrated in Figure 2, adapted from Tijdeman & Seebass (1980), which shows representative instantaneous pressure distributions and their decomposition into amplitude and phase by means of harmonic analysis. Unsteady effects also produce a chord-wise distribution of phase angle relative to the flap deflection that is almost linear.

The ultimate interest in this problem lies in the forces and moments that the oscillatory pressure distribution creates. The moment on the flap lags the flap motion, and once again, single-degree-of-freedom flutter does not occur within the framework of linear theory. However, flap motion *combined* with pitching or plunging motion can easily lead to negative

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aerodynamic damping and catastrophic flutter. Also, as we shall see in subsequent sections, moving shock waves in transonic flow can drastically alter the phase of the hinge moment, leading to the single-degree-of-freedom phenomenon known as "aileron buzz."

Compressibility Effects

Oscillating airfoils at subsonic Mach numbers are not fundamentally different from their incompressible counterparts described in the previous sections, but the solutions are much more difficult to express analytically. Numerical solutions of the linearized potential equation abound, and a series of new analyses in the mid-1970s essentially put the finishing touches on the theoretical understanding of this problem (see Amiet 1975 and Kemp & Homicz 1976, for example). Experiments have provided adequate verification of the two-dimensional theory for small-amplitude oscillations at low and moderate reduced frequencies. However, it should be mentioned that three-dimensional effects on wings become increasingly important as M_∞ increases.

The general form of Equations (1)–(4) is preserved for subcritical Mach numbers, that is, when the flow remains subsonic everywhere. The most important quantitative details are that the pressure, lift, and moment scale inversely with the Prandtl-Glauert factor $\beta = (1 - M_\infty^2)^{1/2}$, as expected

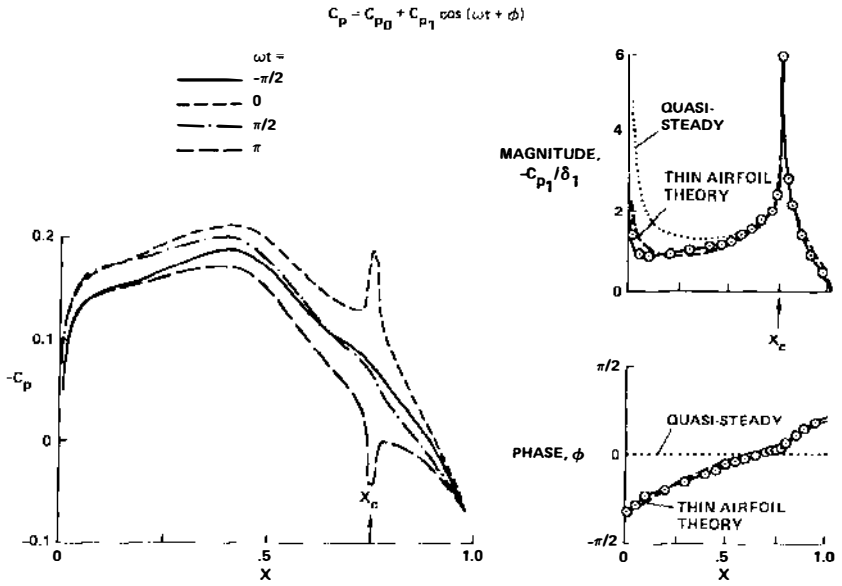


Figure 2 Pressure distributions on an airfoil with an oscillating flap: $M_\infty = 0.5$, $k = 0.39$. —o— experiment.

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from classical steady theory, and the relevant reduced-frequency parameter becomes $k^* = k/\beta^2$. Then the functions $f_n(k^*)$ are relatively insensitive to M_∞ at small values of the quantity $k^* M_\infty$, but the g_n's are generally less so. The curves in Figure 3 show some representative examples, kindly supplied by S. R. Bland in a private communication.

The pressure distribution in *supersonic* flow is, of course, altogether different, and this significantly alters the lift curve slope $dC_L/d\alpha$ and the location of the aerodynamic center. The frequency parameter $\hat{k} = kM_\infty^2/\beta^2$ plays a role similar to k^* in defining unsteady functions $f_n + ig_n$ that multiply the corresponding steady-state solutions. Generally speaking, f_n and g_n vary less with reduced frequency and Mach number than their subsonic counterparts. In both cases, if M_∞ is close to one, linear theory predicts significant reductions in aerodynamic damping, and single-degree-of-freedom flutter can occur for pitch axes near the leading edge (see Lambourne 1968). These cases border on the transonic regime, however, where nonlinear equations have to be used.

In the hypersonic limit, Newtonian flow theory yields approximate results (see Hui & Tobak 1981, for example). Important effects of leading-edge bluntness and surface curvature arise as $M_\infty \to \infty$, including negative

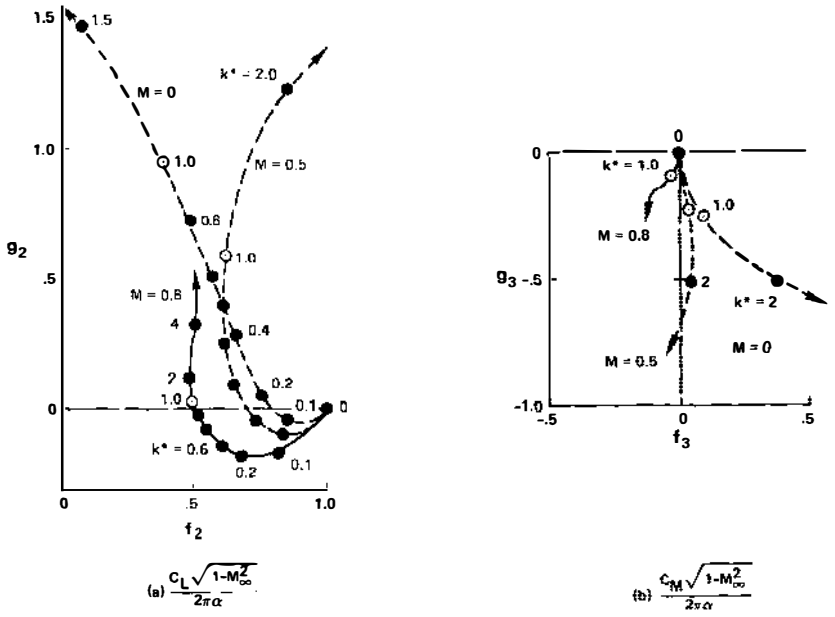


Figure 3 Real and imaginary components of unsteady lift and moment for pitch oscillations: $A = 1/4$.

aerodynamic pitch damping on thin airfoils with sharp leading edges and large concave curvature.

REAL AIRFOILS IN REAL FLUIDS

Modern developments in unsteady-airfoil research can be viewed essentially as attempts to relax the various assumptions that underlie thin-airfoil theory. We first consider effects that are relatively straightforward perturbations of the linear solutions described in the previous section; then we examine highly nonlinear phenomena that drastically alter the nature of the flow.

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Second-Order Effects

Numerous authors have analyzed various complicating factors that can exist on real airfoils, such as airfoil thickness and camber, finite mean angle of attack, large-amplitude motion, boundary-layer displacement thickness effects, and wind-tunnel wall corrections. A few representative examples include the early theoretical analyses of thickness effects by Küssner (1960) and van de Vooren & van de Vel (1964); the finite-difference programs of Giesing (1968) and Desopper (1981) for arbitrary geometry and airfoil motion, the latter including boundary-layer effects; the superposition methods of McCroskey (1973) and Garner (1975); the second-order perturbation analysis of the interaction between a thick cambered airfoil and a gust field by Hamad & Atassi (1980); and the analysis of unsteady wind-tunnel interference by Fromme & Golberg (1980). One general trend that emerges from these and many other studies is that the individual surface pressures are usually affected much more than the *difference* in pressure across the airfoil, which produces the fluctuating lift and pitching moment. It may be mentioned, moreover, that the various approaches generally agree in their predictions of the *trends* of C_P, but there are discrepancies in the predictions of the relatively smaller deviations of C_L and C_M from thin-airfoil theory.

In the linear theory, boundary conditions on the airfoil and in the wake are applied either along the x-axis or along the chordline of the airfoil and a mean streamline in the wake. Provided this mean surface approximation is not violated too strenuously, the various effects classified herein as secondary can be described rather well by superposition of individual corrections to the thin-airfoil solutions discussed previously. This is illustrated in Figure 4, where the superposition method of McCroskey (1973) is applied to the experimental results of McCroskey & Pucci (1981) for a 17%-thick airfoil with moderate camber at $M_\infty = 0.3$. Here the motion is given by $\alpha = \alpha_0 + \alpha_1 \cos \omega t$, where $\alpha_0 = \alpha_1 = 5°$, and the pressure

coefficient can be decomposed into mean, first, and second harmonic components as follows:

$$C_P = C_{P_0} + C_{P_{1c}} \cos \omega t + C_{P_{1s}} \sin \omega t + C_{P_2} \cos(2\omega t + \phi_2) . \quad (5)$$

Also shown in Figure 4 is the flat plate result, Equation (1), which is clearly unsatisfactory for C_P; nevertheless, Δp across the airfoil and consequently C_L (not shown) are given reasonably well by Equations (1) and (2), respectively. For this example the best results were obtained by superposing the individual inviscid contributions (thickness, camber, mean angle, and amplitude) to the surface velocity, correcting for boundary-layer displacement thickness effects, and then using the nonlinear Bernoulli equation to calculate the theoretical unsteady pressure coefficient. Linear theory was used to correct the data for wind-tunnel interference.

What can be said about the results of these refinements? The answer depends on what information is required. Obviously, the details of the individual pressure distributions in Figure 4 are improved, but in this and many other practical cases, thin-airfoil theory gives adequate results for the overall forces and moments. This is somewhat analogous to the steady

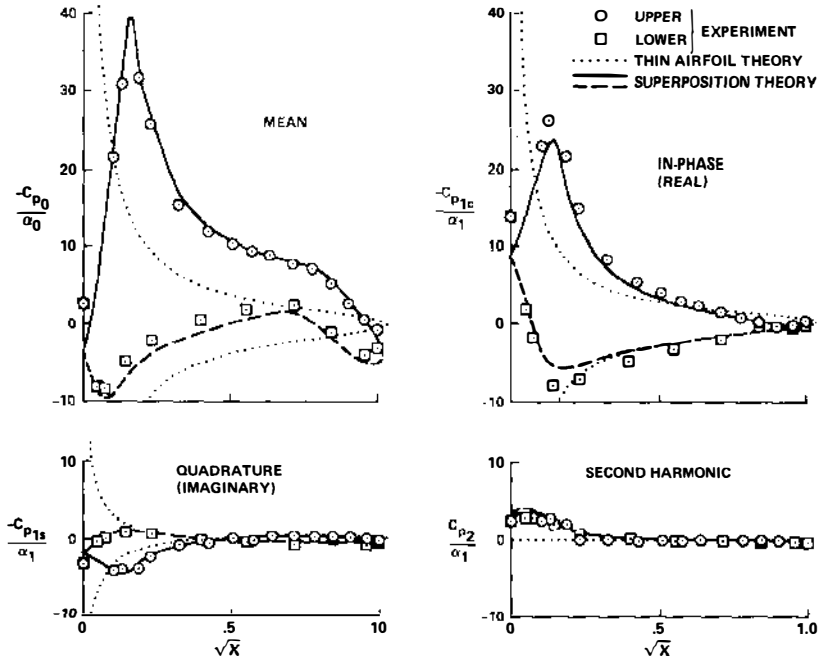


Figure 4 Pressure distributions on an oscillating airfoil: NLR-7301 airfoil, $\alpha = 5° + 5° \cos \omega t$.

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case, in which thickness effects tend to be approximately cancelled by viscous effects in the absence of separation or strong shock waves. However, if boundary-layer calculations are to be done, this requires a realistic pressure distribution. Therefore, meaningful refinements to linearized thin-airfoil theory require that the actual airfoil geometry *and* viscous effects be considered together. The near-term trends are probably best exemplified by Desopper (1981); an unsteady integral boundary-layer method utilizing a quasi-steady turbulence model is coupled to a potential-flow numerical code, at little additional computational expense. This approach is probably adequate so long as no boundary-layer separation occurs.

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Trailing-Edge and Wake Effects

For uniqueness, inviscid airfoil theory invokes the so-called Kutta condition of regular flow in the neighborhood of a sharp trailing edge. In the steady case, this condition is variously interpreted as fixing the rear stagnation point at the trailing edge, avoiding infinite velocities there, or establishing pressure or velocity continuity (or both) across the trailing edge. Refinements that correct for boundary-layer interactions in the trailing-edge region and for wake thickness and curvature effects have been reviewed recently by Melnik (1980), but comparable extensions to the unsteady case remain to be done.

In the more complex unsteady case, the trailing-edge flow situation is poorly understood, physically and mathematically. The usual assumption in inviscid analyses is that no pressure difference can be sustained across the trailing edge and the wake, as illustrated in the upper half of Figure 5. In this model, attributed to E. C. Maskell by Basu & Hancock (1978), the streamlines adjacent to the airfoil leave tangentially to either the upper or lower surface, according to the sign of $d\Gamma/dt$. Furthermore, the wake is usually assumed to remain thin and straight, or at least to follow a steady-flow streamline.

Numerous flow visualizations on oscillating airfoils indicate that the wake has a strong tendency to organize itself into a series of vortices, as indicated in the lower half of Figure 5 and discussed by McCroskey (1977). Few unsteady-airfoil theories include this phenomenon, but more disturbing from the standpoint of theoretical modeling are the experimental indications that finite pressure loading and abrupt streamline curvature can exist in the trailing-edge region under some conditions. Also, the phase lag of the absolute pressure near the trailing edge can deviate significantly from linear theory (see Commerford & Carta 1974 and Lorber & Covert 1981, for example).

Notwithstanding the recent analyses by Daniels (1978) and Yates (1978), a theoretical understanding of this problem is lacking, the exper-

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imental evidence on the range of validity of the Kutta condition is mixed, and the dominant parameters have not been clearly identified. For example, the breakdown of the Kutta-Joukowski condition generally seems to occur as the reduced frequency increases, more so for blunt trailing edges than for sharp (see McCroskey 1977 and Brooks & Hodgson 1979). However, Satyanarayana & Davis (1978) reported the breakdown for $k \simeq 1$ for a thin airfoil oscillating in pitch, whereas Fleeter (1980) concluded that the zero-loading condition prevailed to k in excess of 7 on a flat plate but not for a highly cambered blade in a cascade. Other possibly significant parameters, such as amplitude and type of model motion or flow oscillation, have not been studied systematically.

Further detailed studies are necessary to clarify the nature of the trailing-edge conditions, with or without moderate amounts of trailing-edge separation. The correct theoretical modeling of this region is important in determining the acoustic radiation from wings and rotating blades, in predicting hydrofoil flutter, in analyzing the flow in turbomachinery, in predicting trailing-edge stall on oscillating airfoils, and in understanding certain types of bird and insect flight.

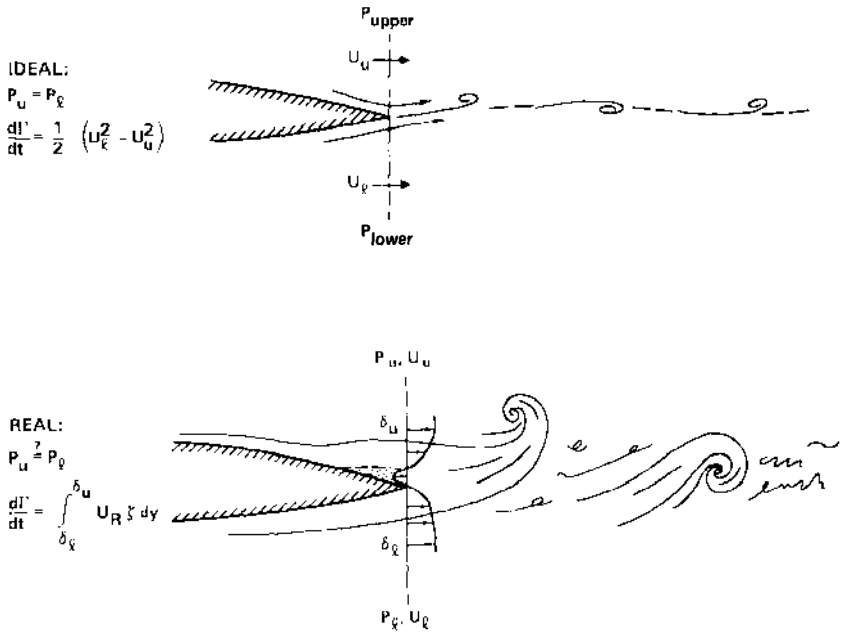


Figure 5 Sketches of some possible trailing-edge conditions in unsteady flow; ζ is the vorticity convected into the wake at relative velocity U_R.

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UNSTEADY TRANSONIC FLOWS

The highly nonlinear flows in this category contain supersonic zones embedded in subsonic fields. The supersonic regions on the airfoil usually terminate with shock waves, whose position and strength are sensitive to small changes in the flow conditions or airfoil parameters. The airloads are much more complicated, and flutter in either single or multiple degrees of freedom is much more likely to occur in the transonic regime than for flows at M_∞ significantly different from unity.

A very large number of papers on unsteady transonic flow has appeared in the recent literature, including a lucid and comprehensive review by Tijdeman & Seebass (1980). Mehta & Lomax (1981) and Nixon (1981) have reviewed computational aspects, Yates & Olsen (1980) have discussed recent experiments, and Borland (1979) has prepared a bibliography that lists more than 200 publications between 1970 and 1979. Since then, over 70 more papers have appeared. With this much recent information readily available, we restrict this section to a brief discussion of the distinguishing features of unsteady transonic flows and to the general trends of current research.

Although it is very difficult to generalize about transonic flows, some idea of the differences from the linear regimes discussed earlier can be obtained from Figure 6. The flow-field development does not pass uniformly from the purely subsonic to completely supersonic regimes; instead, a wide range of patterns can occur in the intermediate stages. Figure 6 merely indicates some representative ones.

As M_∞ is increased above the critical value, shock waves normally form near the middle of the airfoil (Figure 6*b*) and move fore and aft as the airfoil oscillates. In the simplest case, the shock-wave motion is approximately sinusoidal, although it lags the airfoil motion appreciably (see Tijdeman & Seebass 1980, for example). Points on the airfoil that are traversed by the shock wave alternately experience the low pressure ahead of it and the higher pressure behind it. It is important to realize that at any given time, the instantaneous pressure distribution qualitatively resembles the plot of C_{P_0}; but the harmonic components of $C_P(t)$ reflect the large fluctuations due to the moving shock wave, as illustrated by the peaks in the curves of C_{P_1} in Figure 6*b*. This relatively concentrated apparent pressure disturbance, sometimes called a shock doublet, is centered about the mean shock-wave position, and it is a distinguishing feature of periodic transonic flows.

As $M_\infty \to 1$, the shock waves tend to move to the trailing edge, as in Figure 6c. A fishtail pattern of shock waves develops, and the shock doublet tends to disappear. At supersonic speeds (Figure 6d) the pressure coeffi-

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cients, especially the unsteady components, are more nearly constant over the airfoil.

Other and more complex flow patterns are possible in the transonic regime, depending on the airfoil parameters, such as mean angle, geometry, and flap configuration; the parameters of the motion, such as amplitude, reduced frequency, and pitch axis; and an interrelated aspect, the extent and severity of shock-wave and boundary-layer interaction. In the following section, we consider conditions for which the shock-wave motion and mean position are crucial, but unsteady viscous effects are secondary. This is followed by a brief review of unsteady shock-induced separation, which, of course, is much more difficult to analyze and predict.

Nearly Inviscid Flows

In the absence of boundary-layer separation, the motion of the shock wave is the essential feature that makes the unsteady transonic problem nonlinear (see Tijdeman & Seebass 1980, Ashley 1980, and Nixon 1981). The collective experience from theoretical analyses, experiments, and numerical computations indicates that the shock-wave motion is greatest at low to moderate reduced frequencies and in the low transonic regime (Figure 6*b*, for example). The nonlinearity of this regime poses a potentially severe complication when the fluid-dynamic calculations must be coupled to the

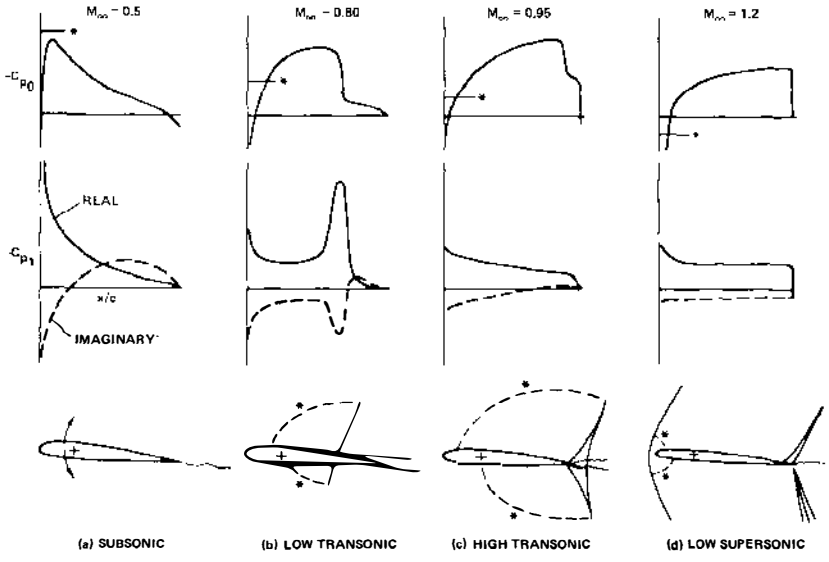


Figure 6 Representative flow fields and pressure distributions at various Mach numbers; * denotes sonic conditions.

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motion of a flexible dynamic structure and many cases analyzed, such as in flutter predictions.

In many cases, however, the most difficult part of the problem is determining the correct *mean* position and strength of the shock wave. The integrated *fluctuating* lift and, to a somewhat lesser extent, the pitching moment, often tend to remain more nearly linear with respect to the motion than do the individual pressures (see Tijdeman & Seebass 1980 and Davis & Malcolm 1980a,b, for example). This is somewhat analogous to the lesser importance of secondary effects with respect to C_L and C_M than to C_P in the purely subsonic domain, discussed previously. Consequently, the superposition principles that are invaluable to aeroelastic analyses tend to remain valid to a surprising extent. When this is true, these analyses can continue to use either harmonic coefficients or indicial functions, even though classical subsonic and supersonic linear theory breaks down and other aerodynamic methods are needed. The requirements for this fortunate situation appear to be that the shock waves not induce boundary-layer separation and that the amplitude of the unsteady motion remain small. However, additional research is needed to establish this more definitively in terms of the airfoil conditions and motion parameters.

If classical linear theory is inadequate, what alternatives are available? Table 1 lists the hierarchy of possible unsteady equations; the equations themselves and detailed discussions of various solution techniques are given by McCroskey (1977), Tijdeman & Seebass (1980), Ashley & Boyd (1980), Nixon (1981), and Mehta & Lomax (1981). Most of the recent and current research activity concerns the small-disturbance equations,

Table 1 Hierarchy of equations for unsteady transonic flow

Equations	Remarks
Navier-Stokes with turbulence modeling	Strong shock waves and separation
Thin-layer Navier-Stokes with turbulence modeling	Strong shock waves, moderate separation
Euler[a]	Inviscid, rotational, strong shock waves
Full potential[a]	Weak shock waves, exact airfoil boundary conditions
Small disturbance	Simplified grids and boundary conditions
Nonlinear[a]	Moderate amplitude
Time linearized[b]	Small amplitude shock motion, harmonic or indicial
Local linearizations[b]	Small amplitudes, harmonic or indicial
Transonic linear theory	High-frequency, stationary shock waves

[a] With or without coupled viscous layer equations.
[b] With or without simplified viscous corrections.

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which may be expressed in many forms (see Ashley & Boyd 1980, for example). For many practical problems, this general level represents a reasonable compromise between acceptable accuracy and the computational efficiency that is required for aeroelastic calculations.

Dowell et al. (1981) recently commented on the suitability of the various equations for engineering applications, based on their study of the boundaries between linear and nonlinear regimes, that is, where superposition is and is not valid for C_L and C_M. Noting that unsteady nonlinear effects diminish with increasing frequency, they suggested seeking the simplest method that would suffice in a given reduced-frequency domain. Figure 7 illustrates this concept for a representative small-amplitude oscillation of an airfoil in a flow with weak shock waves.

In particular, the need for nonlinear unsteady equations might well be limited to the low-frequency region for many applications. For a given airfoil, mean angle, and Mach number, the extent of the nonlinear domain would be expected to diminish with decreasing amplitude of oscillation, and vice versa. Furthermore, Chyu & Schiff (1981) described a technique that could perhaps further reduce the number of nonlinear calculations required. In any case, Dowell et al. (1981) correctly conclude that no method can be useful unless the mean steady flow that it either uses or predicts is correct.

Meanwhile, three other facets of contemporary research are noteworthy. The first is that numerical algorithms and grid-generation techniques for the more accurate full-potential, Euler, and Navier-Stokes equations are

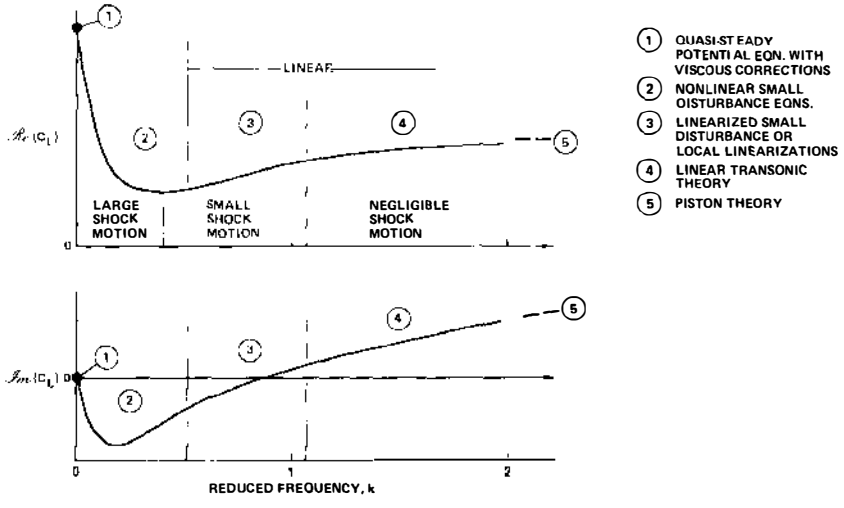


Figure 7 Unsteady flow regimes for a transonic airfoil with weak shock waves.

being vigorously developed for future applications. The second concerns the use of approximations to the far-field boundary conditions that permit the size of the computational domain to be reduced significantly. For example, Kwak (1981) and Fung (1981) have incorporated nonreflecting boundary conditions and linearized far-field solutions, respectively, into small-disturbance codes to accomplish this.

The third important trend is the coupling of viscous effects to various inviscid formulations. Viscous effects can have a major effect on the mean shock-wave position (see Melnik 1980 and LeBalleur 1980 for steady flows), and, in the light of the previous remarks, this should certainly be included. Beyond that, unsteady viscous phenomena are just beginning to be incorporated, but preliminary indications are that they are more important in the transonic than in the subsonic regime. The preliminary studies by Desopper (1981), Couston et al. (1980), and Rizzetta & Yoshihara (1980) presage new results that can be anticipated in the future for relatively mild interactions.

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Strong Viscous Effects

Certain ranges of flow conditions and airfoil-motion parameters produce unsteady shock-induced boundary-layer separation, trailing-edge separation, or various combinations of strong viscous-inviscid interactions that are well beyond the scope of the techniques of the previous section. Some of the more common examples are separation-induced transonic flutter, buffet, and aileron buzz.

Aerodynamic data relevant to flutter problems are normally generated on airfoils undergoing forced oscillations. The experiments of Davis & Malcolm (1980a,b) provide striking examples of the effects of shock-induced separation on the motion of the shock wave and on the pressure loading downstream of the shock. The lift and moment coefficients are also affected dramatically. Accurate predictions of these flows will probably require numerical solutions of the Navier-Stokes equations, but to date this has not been accomplished.

Self-induced flow oscillations, alternating in an approximately regular and periodic manner between trailing-edge and shock-induced separation, have been studied on stationary, biconvex airfoils of 14% and 18% thickness and at zero incidence (see Seegmiller et al. 1978 and Levy 1981). For these cases, numerical solutions of the Reynolds-averaged Navier-Stokes equations were obtained, using an algebraic eddy-viscosity model of the turbulent stresses. These matched the qualitative features of the measured flows and were consistent with the experimental reduced frequencies and Mach-number boundaries for the self-induced oscillations. The review by Mehta & Lomax (1981) may be consulted for additional comments on these investigations.

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Although these biconvex airfoil examples are relevant to flutter, they are actually special cases of the important problem of transonic buffet. The term "buffeting" is used to describe the structural response of aircraft or missile components to the aerodynamic excitation, or "buffet," that is produced by flow separation. In contrast to flutter, the overall body motion is approximately uniform, and the structure normally flexes less under the influence of the buffet airloads.

Buffeting predictions require a knowledge of the fluctuating pressures, which are characterized primarily by the rms pressure coefficient, the dimensionless frequency spectrum, and a frequency parameter $n = \pi kL/c$, where L is a measure of the size of the separated zone and k is the reduced frequency defined previously. Although many experimental results indicate that the spectra may span $0.01 < n < 100$, typically only a small fraction of the total buffet energy is contained above $n \simeq 1$ (see Mabey 1973). This value corresponds very approximately to k of the order of 10 for many typical transonic applications and to frequencies of the same order but somewhat less than the mean frequencies of turbulent eddies quoted by Chapman (1979), assuming L to be of the order of 10 times the viscous layer thickness.

The conditions for the onset of buffet and the spectra of the pressure fluctuations are normally obtained experimentally. However, Mehta & Lomax (1981) have reported on thin-layer, Reynolds-averaged Navier-Stokes calculations by H. E. Bailey and L. L. Levy, Jr., in which buffet-onset boundaries were estimated from the unsteady lift behavior. In accord with the general recommendation of Chapman (1979), it would seem that this and similar Navier-Stokes calculations should be continued and extended to examine the extent to which the broad-band frequency *spectra* of the fluctuating pressures can be predicted, using current quasi-steady turbulence models in the unsteady Reynolds-averaged equations, and whether relatively straightforward extensions can produce useful quantitative results.

The final topic of this section concerns aileron buzz. Aerodynamically excited oscillations of control surfaces can occur when the shock-wave motion and the associated shock doublet introduce negative aerodynamic damping into the hinge moment (see Lambourne 1968 for an excellent discussion of this problem). Although strong viscous effects are not necessarily essential to initiate buzz, in most cases they appear to be crucial in order to sustain its limit-cycle behavior. The most successful analysis of the phenomenon to date is the series of thin-layer Navier-Stokes calculations by Steger & Bailey (1980). Figure 8 illustrates their attempts to predict the measured buzz boundary on a P-80 wing. The numerical simulation of this long-standing problem was not perfect, but the results are unique and quite encouraging.

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It is interesting to note that the extent of the separated regions in Steger & Bailey's results was rather limited. This raises the question of whether comparable results might be obtained from simpler sets of equations that also couple the nonlinear inviscid flow with a thin viscous layer. Based on Lambourne's observations (1968), there are probably a number of buzz conditions for which other zonal modeling, as well as the full Navier-Stokes approach, should be explored.

DYNAMIC STALL AND STALL FLUTTER

A certain degree of unsteadiness always accompanies the flow over an airfoil or other streamlined lifting surface at high angle of attack, but the stall of a thin body undergoing unsteady motion is even more complex than static stall. If the angle of attack oscillates around a mean value α_0 that is of the order of the static-stall angle, large hystereses develop in the fluid-dynamic forces and moments with respect to the instantaneous angle $\alpha(t)$. The maximum values of the lift, drag, and pitching-moment coefficients can greatly exceed their static counterparts, and not even the qualitative behavior of C_L, C_D, and C_M can be reproduced by neglecting the unsteady motion of the airfoil.

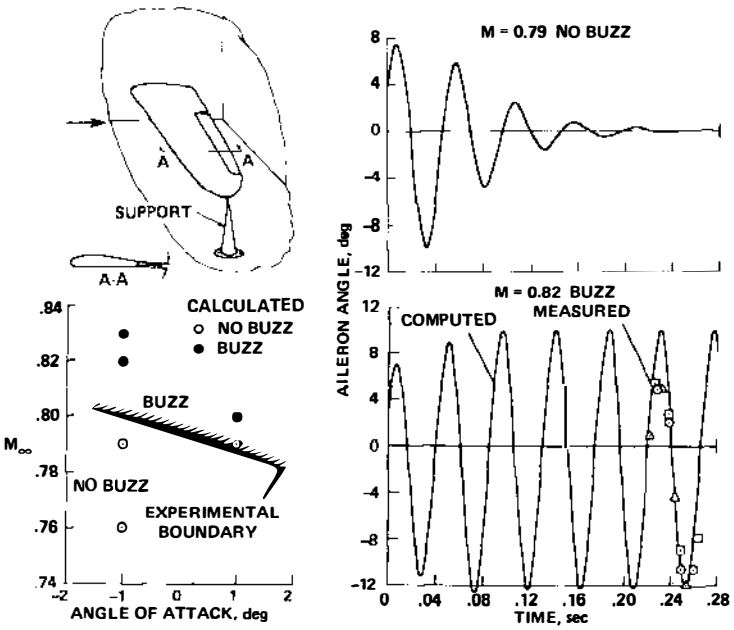


Figure 8 Comparison of measurements and calculations for transonic aileron buzz.

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In addition, the instantaneous aerodynamic damping can become negative during a portion of the cycle. This is illustrated by the dotted shading in Figure 9, adapted from McCroskey & Pucci (1981). If the net damping over the cycle $\zeta = -\oint C_M d\alpha$ is negative, the airfoil extracts energy from the flow, and the pitch oscillations will tend to increase in amplitude, unless restrained. This, of course, is the condition for flutter, and unlike the linear domain described in the first part of this review, the hysteresis in the unsteady separation and reattachment permits flutter to occur in a single degree of freedom of oscillatory body motion.

In practical applications, a distinction is sometimes made between stall flutter and dynamic stall. Stall flutter refers to oscillations of an elastic body that are caused by separated flow which would be nominally steady in the absence of any motion of the body, but which is made unsteady by the flow-induced body oscillations. The term dynamic stall, on the other hand, usually refers to unsteady separation and stall phenomena on airfoils that are forced to execute time-dependent motion, oscillatory or otherwise, or to cases where flow-field perturbations induce transitory stall. Stall flutter and dynamic stall share many common features; the primary fluid-dynamic difference is that the amplitude of the motion is often smaller in stall flutter.

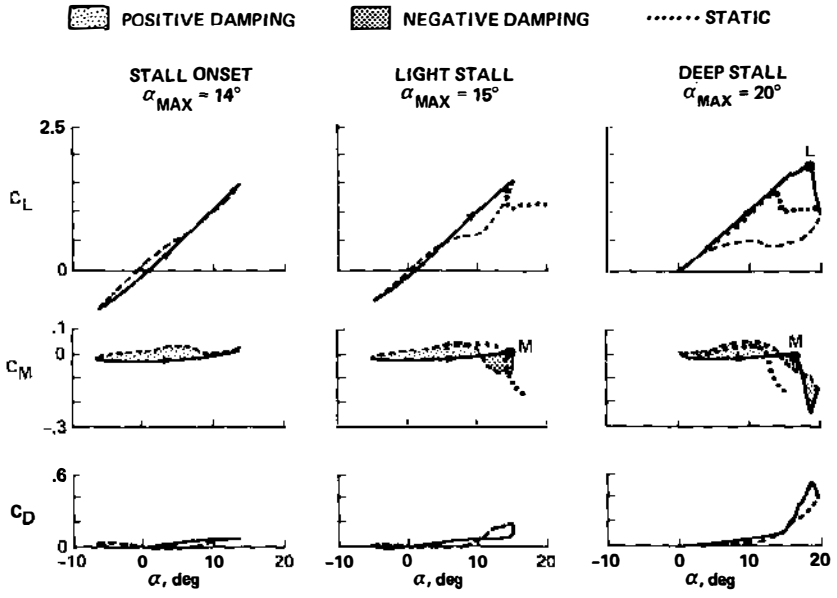


Figure 9 Unsteady forces and moments in three dynamic-stall regimes: $M_\infty = 0.3$, $\alpha = \alpha_0 + 10° \cos \omega t$, $k = 0.10$.

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Stall Regimes

One of the reasons that dynamic stall is more difficult to analyze and predict than static stall is its dependence on a much larger number of parameters. Table 2 lists the more important ones and gives some idea of their significance in delineating the distinctions between static and dynamic stall. Experiments in the last few years have shown that the flow field around an oscillating airfoil in subsonic flow can be characterized by the degree or extent of flow separation and that for a given airfoil, Mach number, and Reynolds number, the primary parameter that determines the degree of separation is the maximum angle of attack ($\alpha_{max} = \alpha_0 + \alpha_1$ for sinusoidal oscillations). This contrasts with the hierarchy of viscous effects on oscillating airfoils at transonic speeds and low angles of attack, where the scale of the interaction is governed primarily by the strength and motion of the shock wave. So far, prediction methods for this class of problems have not been successful for the low-speed, high-angle problems, nor have prediction methods for low-speed, high-angle problems been successful for transonic flows.

The importance of α_{max} is illustrated in Figure 9, which portrays three important regimes of viscous-inviscid interaction for oscillating airfoils. In this particular instance, very little separation occurred when α_{max} was 13° or less, although some viscous effects were noted. When α_{max} was increased to 14°, a limited amount of separation occurred during a small fraction of the cycle and distorted the hysteresis loops of the unsteady pressures and airloads (Figure 9*a*). From a practical standpoint, the effect on C_M and C_D is particularly important. This stall-onset condition represents the limiting case of the maximum unsteady lift that can be obtained with no significant penalty in pitching moment or drag. Further increases in α_{max} produce the dynamic-stall conditions described below, which are charac-

Table 2 Importance of the dynamic-stall parameters

Stall parameter	Effect
Airfoil shape	Large in some cases
Mach number	Small below $M_\infty \sim 0.2$; large above $M_\infty \sim 0.2$
Reynolds number	Small (?) at low Mach number; unknown at high Mach number
Reduced frequency	Large
Mean angle, amplitude	Large
Type of motion	Virtually unknown
3-D effects	Virtually unknown
Tunnel effects	Virtually unknown

terized by large phase lags and hystereses in the separation and reattachment of the viscous flow.

LIGHT STALL Figure 9*b* illustrates the next level of viscous-inviscid interaction, which was obtained at a slightly greater value of α_{max} with all other conditions remaining the same. This category of dynamic stall shares some of the general features of classical static stall, such as loss of lift and significant increases in drag and nose-down pitching moment compared with the theoretical inviscid values, when α exceeds a certain critical value. However, the *unsteady* stall behavior is characterized by growing hysteresis in the airloads. Also, the tendency toward negative aerodynamic damping, as discussed above, is strongest in this regime.

Another distinguishing feature of light dynamic stall is the scale of the interaction. The vertical extent of the viscous zone tends to remain of the order of the airfoil thickness, illustrated in Figure 10*a*, and this is generally less than for static stall. Consequently, this class of oscillating-airfoil problems should be more within the scope of zonal methods or thin-layer Navier-Stokes calculations with relatively straightforward turbulence modeling than either static stall or deep dynamic stall.

The qualitative behavior of light stall is known to be especially sensitive to airfoil geometry, reduced frequency, maximum incidence, and Mach number; also, three-dimensional effects and the type of motion are probably important. The quantitative behavior is closely related to the boundary-layer separation characteristics, for example, leading-edge versus trailing-edge separation, and to the changes in this separation behavior with α_{max}, k, and M_∞. The effects of these parameters are described in more detail by McCroskey (1981) and McCroskey et al. (1981).

DEEP STALL Figure 9*c* shows the effects of increasing the maximum incidence to values well in excess of the static-stall angle. The initial breakdown of the flow in the deep-stall regime begins with the formation of a strong vortex-like disturbance in the leading-edge region. This vortex is shed from the boundary layer and moves downstream over the upper surface of the airfoil, producing values of C_L, C_M, and C_D that are far in excess of their static counterparts when α is increasing; large amounts of hysteresis occur during the rest of the cycle. The scale of the viscous interaction zone is also large; the thickness of the viscous layer is of the order of the airfoil chord during the vortex-shedding process, as illustrated in Figure 10*b*.

Figures 9 and 10 illustrate some of the qualitative and quantitative differences in light and deep dynamic stall. Moment stall, denoted by M in Figure 9, occurs rather abruptly in both cases, but the deep-stall drop in lift after $C_{L_{max}}$, denoted by L, is not evident in the light-stall case. The large negative values of C_M in deep stall are due to the vortex.

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A combination of large amplitudes of the oscillation (or large pitch rates in the case of nonperiodic motion) *and* large maximum angles is essential to the development of deep stall. If these conditions are fulfilled, and provided that M_∞ is low enough that leading-edge shock waves do not develop, then the qualitative features are relatively insensitive to the details of the airfoil motion, airfoil geometry, Reynolds number, and Mach number. However, Mach number becomes an important parameter if the flow becomes supersonic in the leading-edge region. The quantitative airloads depend primarily on the time history of the angle of attack for the portion of the cycle when α exceeds the static stall angle, α_{ss}. Again, this feature and other details of the flow behavior are discussed at greater length by McCroskey (1977, 1981) and McCroskey et al. (1981).

Methods of Calculation

Theoretical progress remains rather slow on this difficult problem, although a number of prediction methods have been developed and are still

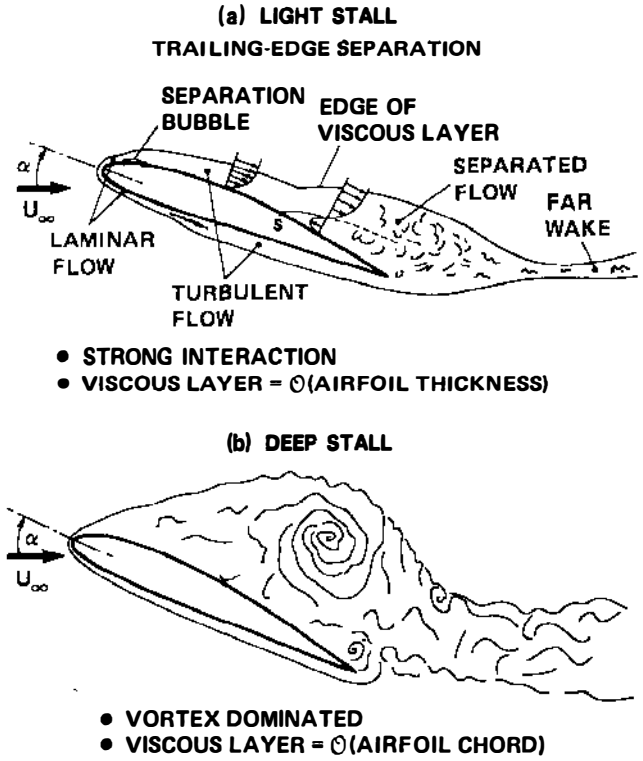


Figure 10 Sketches of flow fields during dynamic stall.

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being refined. The techniques that exist all invoke restrictive assumptions and approximations, and they are tailored to the specific features of some particular stall regime or to a relatively narrow range of parameters. The brief descriptions below are condensed from the recent review by McCroskey (1981) which also cites the original references for the various approaches.

The organized vortex-shedding phenomenon of the deep-stall regime has motivated several attempts to model it by discrete potential vortices, analogous to the vortex methods for bluff-body flows. The crux of this general approach lies in choosing the strength and location of the vortex emissions, and in relating their properties to the boundary-layer separation characteristics on the body. Also, computational expense limits the time step size and the total number of vortices in the flow field. The qualitative features of deep stall have been simulated well by this approach, but further refinements are required for suitable engineering accuracy at reasonable computational efficiency.

Zonal methods of viscous-inviscid interaction, similar to those mentioned in the previous section, are being developed and refined in several laboratories, but no entirely satisfactory dynamic-stall predictions are available thus far. The same can be said of numerical solutions to the Reynolds-averaged Navier-Stokes equations for high-Reynolds-number flows, although the recent calculations of Tassa & Sankar (1981) are encouraging and several laminar calculations have been published (see McCroskey 1977). An increasing output of both zonal methods and Navier-Stokes calculations can be expected in the next few years.

The helicopter industry has developed several engineering prediction techniques based on empirical correlations of wind-tunnel data for estimating the unsteady airloads on oscillating airfoils. These methods seek to correlate force and moment data obtained from relatively simple wind-tunnel tests as functions of the numerous relevant parameters, such as airfoil shape, Mach number, amplitude and frequency of oscillation, mean angle, and type of motion.

Common to all the available literature relevant to dynamic stall is the observation that unsteady effects increase with increasing pitch rate, that is, with rate of change of airfoil incidence. It is also evident that the dynamic-stall events require finite times to develop. Therefore, some form of the nondimensional parameters $\dot{\alpha}c/U_\infty$ and $U_\infty \Delta t/c$ appears in all of the empirical methods. Another common aspect is that the empirical correlations are used as corrections to steady-airfoil data, so that the geometrical, Reynolds-number, and Mach-number effects are included only insofar as they determine the static section characteristics.

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The most recent method to come to the author's attention is that of Tran & Petot (1980). In contrast to the earlier methods that essentially curve-fit experimental data with various algebraic or transcendental functions, this method utilizes a system of ordinary differential equations. Evaluating the empirical coefficients requires a large quantity of small-amplitude data over a wide range of reduced frequencies, but a systematic way of generating these data very efficiently has been developed. The results obtained so far have been encouraging, but the method fails to treat large-amplitude deep-stall cases adequately (e.g. Figure 9*c*). More satisfactory results from this and the other empirical-correlation techniques await a better theoretical understanding of the basic separation and reattachment phenomena on oscillating airfoils.

CONCLUDING REMARKS

As we have seen, the major thrusts in unsteady-airfoil research are directed at understanding and predicting nonlinear phenomena that are beyond the scope of classical thin-airfoil theory. The two main classes of unsteady-flow separation indicated in Figure 11 represent the greatest challenges for future research and development.

Considerable progress has been made since the author's 1977 review in developing numerical-solution techniques for the compressible Navier-Stokes equations, which are required for complex transonic flows with extensive separation. Further refinements are needed, of course, but with this new capability it would seem that the problem of buffet airloads should be addressed next.

On the other hand, a wide range of approximate techniques and efficient numerical algorithms have become available for the simpler cases that can

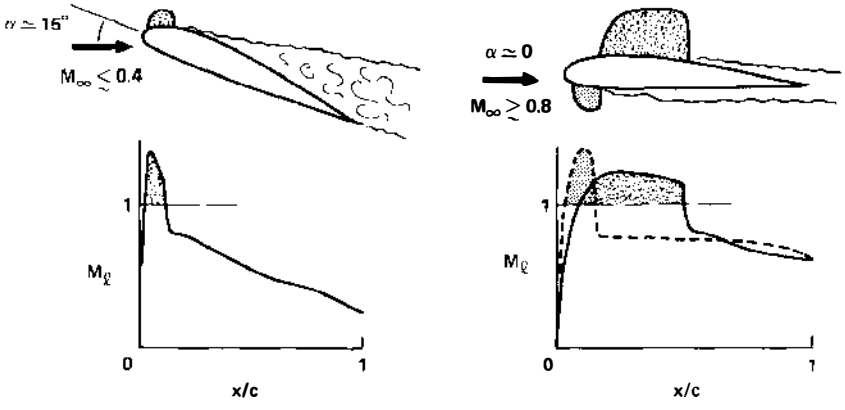


Figure 11 The twin challenges of oscillating-airfoil research.

be treated with the inviscid, transonic small-disturbance equations. For more complex flows with limited separation, vigorous development of zonal methods that couple thin viscous-layer equations with the full-potential or Euler equations seems essential and, in fact, inevitable.

Progress has lagged for flows at high angles of attack, with or without shock waves. Thus far, experiments have provided most of the fundamental knowledge and quantitative information, and the viscous behavior in the presence of large leading-edge pressure gradients has not been analyzed satisfactorily. Additional complicating factors that are important in this case are the transition and early stages of turbulent-flow development in the leading-edge region.

The qualitative features of dynamic stall are incorporated in the current empirical prediction methods that have been developed for helicopter applications, but all of them need further improvements. The recent identification of the distinct features of light and deep dynamic stall provides clearer targets for the different analytical and numerical approaches that are currently under development. Therefore, we can expect more rapid progress on this problem in the next few years.

Finally, we must appreciate the limitations of the two-dimensional viewpoint taken in this review. Useful concepts and theoretical models of unsteady phenomena can be obtained from such an approximation, and many engineering prediction methods are based on judicious applications of "strip theory" and nominally two-dimensional data. However, almost all real problems are three-dimensional to some extent, and there are often significant quantitative and occasionally major qualitative differences between two- and three-dimensional unsteady effects. Therefore, the main thrusts of unsteady fluid dynamics must ultimately be directed to three dimensions.

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